Visualizing these numbers on the number line, we see that



Fig. 1.7

-8 is to the right of -27 and hence $(-8) > (-27)$. This leads to the result that $\frac{-8}{12} > \frac{-27}{12}$ and consequently we conclude that $\frac{-2}{3} > \frac{9}{-4}$.

Example 1.7

Write the following rational numbers in ascending and descending order.

$$\frac{-3}{5}, \frac{7}{-10}, \frac{-15}{20}, \frac{14}{-30}, \frac{-8}{15}$$

Solution:

First make the denominators to be positive and write the numbers in standard form as $\frac{-3}{5}, \frac{-7}{10}, \frac{-15}{20}, \frac{-14}{30}, \frac{-8}{15}$. Here, the LCM of 5,10,15,20 and 30 is 60 (Find it!). Change the given rational numbers in equivalent form with common denominator 60.

$$\frac{-3}{5} = \frac{-3}{5} \times \frac{12}{12} = \frac{-36}{60} \qquad \frac{-7}{10} = \frac{-7}{10} \times \frac{6}{6} = \frac{-42}{60}$$

$$\frac{-15}{20} = \frac{-15}{20} \times \frac{3}{3} = \frac{-45}{60} \qquad \frac{-14}{30} = \frac{-14}{30} \times \frac{2}{2} = \frac{-28}{60}$$

$$\frac{-8}{15} = \frac{-8}{15} \times \frac{4}{4} = \frac{-32}{60}$$

Comparing the numerators alone, that is, $-36, -42, -45, -28$ and -32 we see that

$$-45 < -42 < -36 < -32 < -28$$

Hence, $\frac{-45}{60} < \frac{-42}{60} < \frac{-36}{60} < \frac{-32}{60} < \frac{-28}{60}$ and so, $\frac{-15}{20} < \frac{7}{-10} < \frac{-3}{5} < \frac{-8}{15} < \frac{14}{-30}$.

So, the ascending order is $\frac{-15}{20}, \frac{7}{-10}, \frac{-3}{5}, \frac{-8}{15}$ and $\frac{14}{-30}$.

Also, its reverse order gives the descending order as $\frac{14}{-30}, \frac{-8}{15}, \frac{-3}{5}, \frac{7}{-10}$ and $\frac{-15}{20}$.

1.2.7 Rational numbers between any two given rational numbers

Consider the integers 4 and 10. We can locate five integers namely 5,6,7,8 and 9 (shown in dark dots) between them. Isn't it?



Fig. 1.8

How many integers can you find between 3 and −2? List them.

Are there any integers between −5 and −4? No, is the answer.

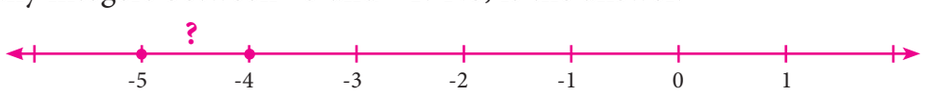


Fig. 1.9

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